Aligning LLMs with Biomedical Knowledge using Balanced Fine-Tuning

Zhenchao Tang*[1], Fang Wang*†[1], Haohuai He[1], Jiale Zhou[1], Tianxu Lv[1], Jun Zhu[1], Shouzhi Chen[1], Minghao Yang[1], Yu Wang[1], Jiayang Wu[1], Yidong Song[2], Yaokun Li[3], Jiehui Huang[4], Jun Zhou[5], Bing He[1], Jianhua Yao†[1]

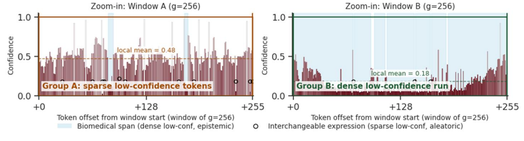
[1]Tencent AI for Life Sciences Lab [2]Nankai University [3]Sun Yat-sen University [4]HKUST [5]Tsinghua University

*Equal contribution †Correspondence: avonwanghit@gmail.com, jianhua.yao@gmail.com

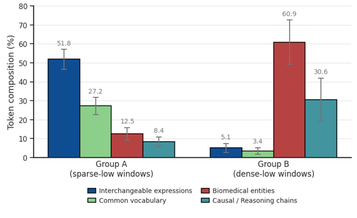
NeurIPS 2026
arxiv.org/abs/2511.21075
shapsider.github.io/BFT

1 Challenge: useful knowledge is difficult

Confidence-based training can suppress unfamiliar biomedical words. An isolated difficult word and a whole difficult passage need different treatment.



2 Motivation: learn difficult passages



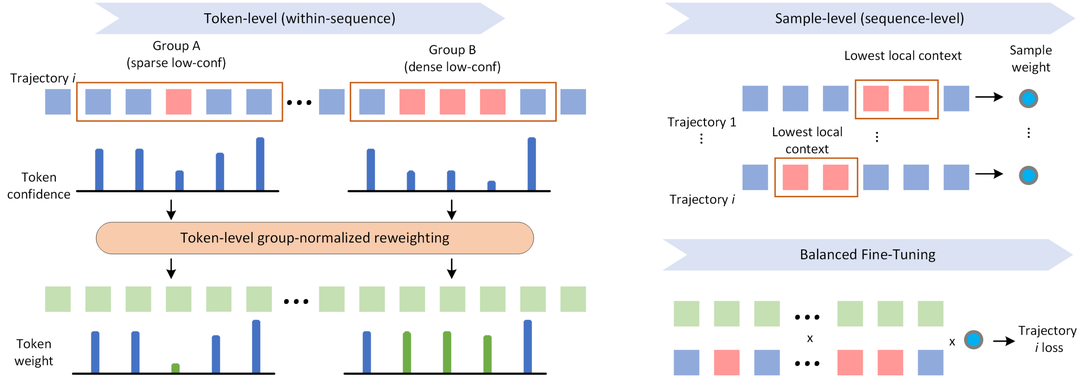
Grouped difficult passages contain more biomedical entities (60.9% vs 12.5%) and reasoning content (30.6% vs 8.4%).

Preserve useful difficulty while retaining broad knowledge.

These associations motivate BFT. They do not identify every uncertain word as useful knowledge.

3 Method: Balanced Fine-Tuning

A simple change to supervised training: use nearby context to decide which words and training examples deserve more learning attention.

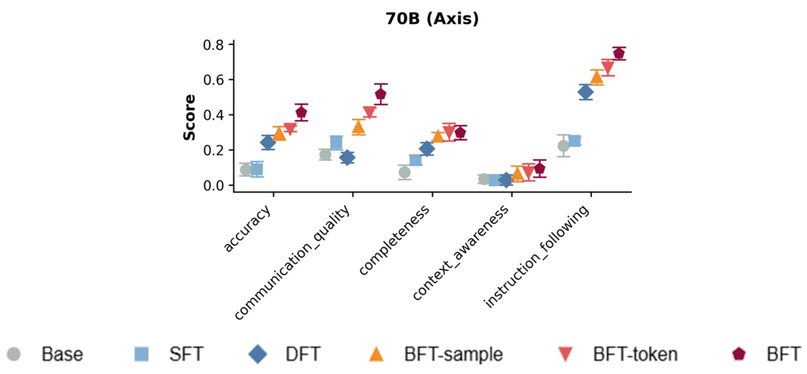


Left: retain words inside difficult passages. Right: emphasize examples containing hard passages, within a limit. BFT uses the model's own confidence, without a reward model or external uncertainty estimator.

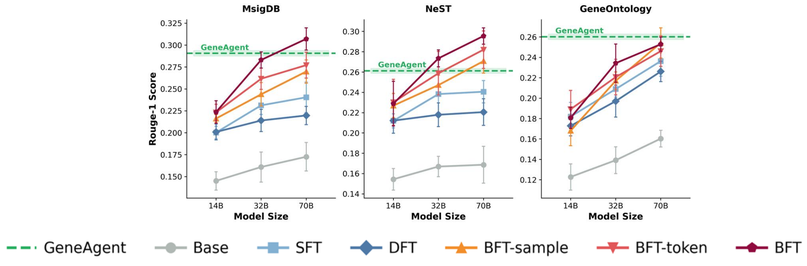
4 Experimental results: biomedical gains with less general forgetting

Stronger biomedical reasoning

HealthBench Hard (70B): medical competence across five evaluation dimensions.

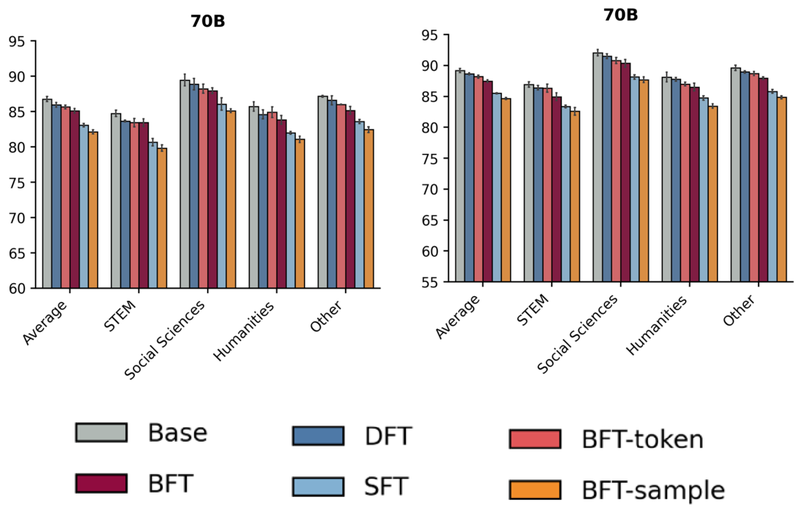


Biological process reasoning



Preserving general capability

MMLU and CMMLU (70B): BFT preserves more general knowledge than SFT while improving biomedical skills.

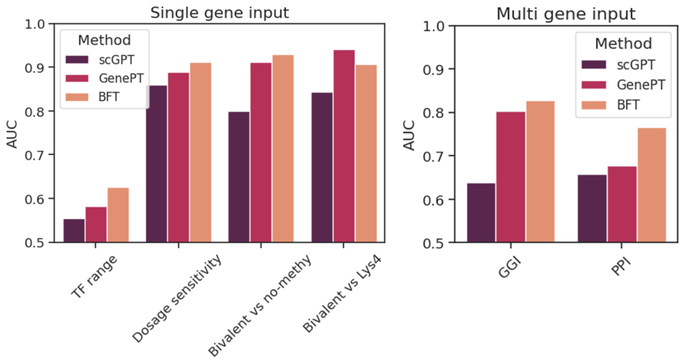


Both benchmarks show less forgetting than SFT and sample-only BFT. Some forgetting remains. DFT forgets least, while BFT offers stronger biomedical adaptation.

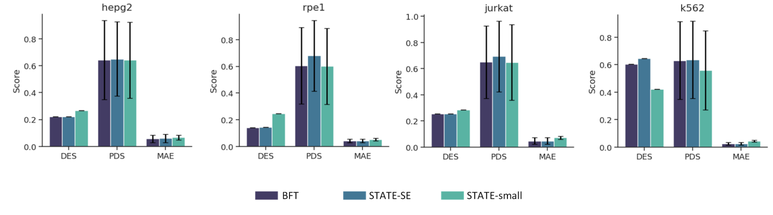
STEM adaptation demonstrated in biomedicine.

Useful scientific representations

Gene descriptions become inputs for gene prediction and single-cell response modeling.



Mean gene score: BFT 0.829 vs GenePT 0.802. GenePT remains higher on Lys4. Both use the same text encoder.



Comparable STATE results across four datasets. Profile generation is zero-shot here; the STATE model is still trained.